

# SAMMY MUSTAFA

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## RESEARCH INTERESTS

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Artificial Intelligence and Machine Learning in Biotechnology, Clinical Trial Informatics, Healthcare Equity and Access, Translational Biomedical Informatics, Digital Health Technologies, Precision Medicine

## EDUCATION

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**Feinberg School of Medicine**

*MD*

Chicago, IL

*Incoming 2026, Expected 2030*

**Harvard Medical School**

*M.S., Biomedical Informatics (GPA: 3.97)*

Boston, MA

*2024*

**Northwestern University**

*B.A., Biological Sciences, Minors in Data Science & Linguistics (GPA 3.93)*

Evanston, IL

*2023*

## PROFESSIONAL EXPERIENCE

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**Bond Health**

*Co-Founder & CIO*

New York, NY

*Jan 2024 – Present*

- Build proprietary LLM-based platform to improve the efficiency and inclusivity of clinical trial matching and patient screening
- Lead regulatory and compliance strategy (HIPAA, BAA agreements, and data security frameworks) for pilots with academic health systems and pharmaceutical sponsors
- Direct technical roadmap for ML-based patient screening and trial-matching pipelines, ensuring adaptability across diverse populations and clinical indications
- Conduct stakeholder interviews with principal investigators, clinical research coordinators, patients, pharmaceutical sponsors, and outsourcing partners (CROs) to guide product design and ensure alignment with industry needs

**Behaivior**

*Data Scientist*

Remote

*Sep 2023 – Dec 2023*

- Applied RNNs and ensemble models on 200,000+ multimodal physiological datapoints to predict craving episodes in substance use recovery patients
- Enhanced unsupervised learning pipelines by integrating label propagation with scikit-learn clustering, achieving a 15% improvement in predictive precision for patient stratification
- Optimized the Cypher algorithm for real-time anomaly detection in wearable device streams, enabling faster intervention alerts and improved patient monitoring
- Contributed to development of digital therapeutic tools combining wearable sensors, ML models, and behavioral interventions to support long-term addiction recovery

## **H2Ok Innovations**

Boston, MA

*Associate Consultant*

*Oct 2023 – Dec 2023*

- Led the implementation of AI-powered, IoT-enabled systems optimizing industrial liquid management, deploying scalable solutions across Fortune 500 clients in manufacturing and utilities
- Served as project manager for strategic market and sensitivity analyses in a potential \$14Bn market segment, identifying key entry points and value propositions
- Boosted revenue potential by 25% by designing bespoke pricing strategies grounded in competitive benchmarking and customer needs analysis
- Collaborated closely with cross-functional teams (data scientists, engineers, and business development) to align technical feasibility with market demand and regulatory considerations

## **Mercurial-AI**

Chicago, IL

*Strategy Consultant*

*Mar 2023 – Aug 2023*

- Spearheaded operational reconfiguration and go-to-market strategy that enabled the company to secure over \$300,000 in seed funding
- Served as project lead for developing a deep-ensemble and causal modeling framework to tailor treatment recommendations for breast cancer patients
- Conducted comprehensive competitive landscape and total addressable market (TAM) analyses, providing strategic insights for AI-enabled oncology product positioning and differentiation
- Prepared investor materials and strategic briefs that translated technical milestones into compelling business cases, strengthening communication with venture capital and clinical collaborators

## **Northwestern University Residential Service**

Evanston, IL

*Resident Assistant*

*Aug 2021 – Jun 2023*

- Oversaw 44 residents, served overnight duty shifts, worked the front desk, and developed community builder events
- Represented the RAs on the RA Advisory Board, advocated for adjustments to the position and services

## **RESEARCH EXPERIENCE**

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### **Center for Precision Psychiatry, Massachusetts General Hospital**

Boston, MA

*ML Researcher*

*Dec 2023 – Dec 2024*

- Developed a metabolomics framework for early detection of major depressive disorder (MDD), stratified by severity and phenotype
- Implemented multivariable logistic regression, mutual information pipelines, and Louvain clustering across 280,000 UK Biobank participants and 250+ metabolite features
- Identified phenotype-specific lipid biomarkers predictive of MDD heterogeneity, informing personalized psychiatry approaches
- Applied ML-based residualization models to separate genetic vs. environmental contributions to metabolic biomarkers

**Potocsnak Longevity Institute, Feinberg School of Medicine**

Chicago, IL

*Molecular Biology Researcher*

Mar 2021 – Jun 2023

- Probed the RNA-binding protein SERBP1 as a regulator of PAI-1, a central mediator of tissue fibrosis, senescence, aging, and Alzheimer's disease
- Designed and executed an independent project using GFP-tagged SERBP1 to study subcellular localization (nucleus vs. cytoplasm) under high glucose and small-molecule inhibitor treatments
- Identified SERBP1's potential prion-like, transgenerational behavior with implications for epigenetic regulation of PAI-1 expression and long-term fibrosis risk
- Proposed therapeutic strategies targeting SERBP1 localization and binding to destabilize PAI-1 mRNA, informing future directions in cardiovascular and aging-related disease intervention

**Behavioral Research Lab, Yale University**

Remote

*NLP Research Associate*

Mar 2022 – Jul 2022

- Conducted comparative analyses of bargaining behaviors in verbal vs. non-verbal negotiation experiments, supporting a large-scale behavioral economics + NLP dataset
- Analyzed over 6,000 minutes of negotiation transcripts using Python, employing natural language processing techniques (sentiment analysis, word embeddings, LIWC features) to quantify and classify linguistic behaviors
- Assisted in annotating thousands of speech acts (pushes, comparisons, allowances, offers), helping establish one of the largest negotiation dialogue corpora to date
- Findings revealed that verbal communication led to faster convergence, higher agreement rates, and reduced variance in negotiated prices

**Complex Systems Lab, Polytechnic Institute of Catalonia**

Barcelona, Spain

*Systems Biology Research Assistant*

Jun 2022 – Sep 2022

- Designed mathematical and spatial resolution models of the cAMP-PDE signaling pathway in atrial myocytes via MATLAB
- Developed differential formulas to model the autonomic effects of atrial fibrillation risk variant rs13143308T
- Investigated the modulation of the PDE1-Adenosine A<sub>2A</sub> macromolecular cluster on sarcoplasmic/cytosolic calcium levels

**PEER-REVIEWED PUBLICATIONS**

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- **Mustafa S**, Moraitaki H, Celi LA, Hao S, Goode D, Kahan T. Evaluating pain assessment and management in acute pancreatitis. *MIT Health and Life Sciences (HEALS) Conference*, Cambridge, MA, December 4, 2024. Preprint: <https://sammymustafa.com/research/heals.pdf>
- Sharon DA, Huang Y, Oyewole M, **Mustafa S**. How to Go With the Flow: An Analysis of Flow Matching Molecular Docking Performance With Priors of Varying Information Content. *ICLR 2024 Workshop on Generative and Experimental Perspectives for Biomolecular Design (GEM)*, Vienna, Austria, May 11, 2024. OpenReview: <https://openreview.net/forum?id=FGwGt9XWO4>

## MANUSCRIPTS UNDER REVIEW

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- **Mustafa S**, Shachar C. A Medico-Legal Approach to Subjective Pain Testimony in Disability Adjudication. Harvard Law School. 2024. Paper: <https://sammymustafa.com/research/hls.pdf>
- Chen Y, **Mustafa S**, Picard R. Integrating Anticipatory and Physiological Factors to Predict Injection Pain Intensity. Affective Computing Lab, MIT Media Lab. 2024. Paper: <https://sammymustafa.com/research/media.pdf>
- **Mustafa S**, Watts D, Smoller JW. Metabolic Biomarkers of MDD Phenotype Heterogeneity. Center for Precision Psychiatry, Massachusetts General Hospital. 2024. Paper: <https://sammymustafa.com/research/psych.pdf>
- Ameen S, Chen X, Liu G, **Mustafa S**, Rigby T. Diagnosing Autism Spectrum Disorder from 3D MRI Images in Pediatric Cases. Department of Biomedical Informatics, Harvard Medical School. 2024. Paper: <https://sammymustafa.com/research/mri.pdf>
- **Mustafa S**, Vaughan D. SERBP1: Exploiting RNA-binding protein-mediated PAI-1 inhibition. Potocsnak Longevity Institute, Northwestern University Feinberg School of Medicine. 2022. Paper: <https://sammymustafa.com/research/serbp1.pdf>
- **Mustafa S**, Leonardi D. PTENP1: A Pivotal Pseudogene in Glioblastomas. Cell Biology Lab, Bergen County Academies. 2020. Paper: <https://sammymustafa.com/research/ptenp1.pdf>

## HONORS/AWARDS/FUNDING

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|---------------------------------------------------------------|------|
| \$150K CEAS SAFE Investment at \$3M Valuation                 | 2025 |
| Harvard Law Entrepreneurship Project Awardee                  | 2025 |
| NSF I-Corps Spark/Fusion Program Grant Awardee (\$1500)       | 2025 |
| MIT Sandbox Innovation Fund Awardee (\$2,500)                 | 2025 |
| MIT 100K Pitch Competition Semifinalist (\$2000)              | 2024 |
| Top 20 Team @ Penn Healthcare Case Competition                | 2023 |
| Weinberg Research Grant (\$4,000)                             | 2022 |
| Honors Program in Medical Education Research Grant (\$3,000)  | 2021 |
| Thermo Fisher Scientific Antibody Scholarship Award (\$5,000) | 2020 |
| Regeneron Science Talent Search Scholar (\$2,000)             | 2020 |

## SERVICE & COMMUNITY OUTREACH

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**Cambridge After School Program (CASP)**  
*After-School Volunteer*

Cambridge, MA  
Jun 2024 – Dec 2024

- Providing homework help, mentorship, and enrichment activities for low-income, immigrant youth in Cambridge (grades 3-8)

### **Lifestyle Interventions for Youth (LIVELY)**

Boston, MA

*Workshop Facilitator*

*May 2024 – Dec 2024*

- Led interactive workshops on lifestyle medicine pillars (nutrition, physical activity, sleep, stress management, substance avoidance, and social connections) for youth ages 13-14, fostering early health education and wellness practices
- Promoted early health education by leading peer discussions that connected physical and mental health, empowering adolescents at the Boston Boys & Girls Clubs to adopt sustainable wellness habits

### **Camp Kesem**

Evanston, IL

*Unit Leader*

*Sep 2022 – Aug 2023*

- Served as Unit Leader at two 5-day camps for children (ages 10-12) whose parents/guardians were affected by cancer, ensuring well-being, safety, and inclusion in activities
- Oversaw medication schedules, daily routines, and emotional support for ~20 campers, fostering resilience and a supportive community environment
- Facilitated group discussions and recreational activities that encouraged peer connection, empowerment, and coping strategies for youth impacted by parental cancer

### **Books & Breakfast (B&B)**

Evanston, IL

*Before-School Tutor & Mentor*

*Sep 2022 – Aug 2023*

- Tutored and mentored elementary students from low-income and immigrant backgrounds to improve academic readiness, literacy, and confidence before the school day
- Supported an equity-focused program providing nutritious breakfasts, 1-1 English tutoring in the summer, and holistic support

### **Evanston Grows**

Evanston, IL

*Community Gardener Volunteer*

*Sep 2022 – Jun 2023*

- Supported community-led initiatives to reduce food insecurity by growing local produce and strengthening neighborhood food distribution networks

## **PROFESSIONAL ASSOCIATIONS**

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|--------------------------------------------------------------------|-----------------------|
| NIH CTSA Recruitment Innovation Center (RIC) Community of Practice | <i>2025 – Present</i> |
| Illinois Alliance to Promote Opportunities for Health (IAPOH)      | <i>2025 – Present</i> |
| Institute of Electrical and Electronics Engineers                  | <i>2024 – Present</i> |